Marco Abbadini

POSTDOCTORAL RESEARCHER

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Highlights

*My research develops **Stone-type categorical dualities** and representation theorems linking the algebraic semantics of logic with topological structures. I combine methods from universal algebra, topology and category theory to study structures arising from logic, including MV-algebras, lattice-ordered groups, hyperdoctrines.*

SELECTED PUBLICATIONS

- Abbadini, Marra, Spada. Stone–Gelfand duality for metrically complete lattice-ordered groups. ***Advances in Mathematics*** (2025).
- Abbadini, Di Liberti. Duality for coalgebras for Vietoris and monadicity. ***The Journal of Symbolic Logic*** (2025).
- Abbadini, Aglianò, Fioravanti. Varieties of MV-monoids and positive MV-algebras. ***Journal of Algebra*** (2025).

MAJOR FUNDING & PROJECTS

- **Marie Skłodowska–Curie** Postdoctoral Fellowship (MSCA PF) (University of Birmingham, UK), 2023–2025.
  Project: *DCPOS: Dualities via ComPactness, Openness and their Symmetry*.
  Supervisor: Achim Jung.
- **FSR Incoming** Postdoctoral Fellowship (Université catholique de Louvain, Belgium), 2025–2028.
  Project: *CatCoD: Categorical Commutators and Dualities*.
  Supervisor: Marino Gran.

HIGHLIGHTS

- Publications: 22 publications, including accepted papers (19 journal articles, 2 book chapters, 1 proceedings paper).
- Invited talks: invited talks, including plenary talks such as *Topology, Algebra, and Categories in Logic 2026* and *BLAST 2023*.
- Teaching: teaching experience from first-year Bachelor's to Master's level, including as course instructor for *Categorical Dualities in Logic* (2025/26); Master's thesis reader and co-supervision of BSc theses.
- Awards: AILA Paolo Gentilini Prize (2026), AILA Ada Lettieri Prize (2025), and special mention for the AILA–UMI Franco Montagna Prize (2023).
- National qualifications: Italian ASN (Seconda Fascia, 01/A1 Mathematical Logic, 2024); French MCF qualification (Mathematics & Computer Science, 2023).

Full publication list, talks, teaching and service appear in the following pages.

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Academic positions and education

Academic positions

FSR Incoming Postdoctoral Fellowship *2025.09.01 – present*

Université catholique de Louvain (Belgium), Research Institute in Mathematics and Physics.

- Project: *CatCoD: Categorical Commutators and Dualities*.
- Supervisor: Marino Gran.

Marie Skłodowska–Curie Postdoctoral Fellowship (MSCA PF) *2023.09.01 – 2025.08.31*

University of Birmingham (UK), School of Computer Science.

- Awarded after EU MSCA evaluation; funded by UK Research and Innovation under the Horizon Europe funding guarantee.
- Project: *DCPOS: Dualities via ComPactness, Openness and their Symmetry*.
- Supervisor: Achim Jung.

PostDoc *2020.12.01 – 2023.08.31*

University of Salerno (Italy), Department of Mathematics.

- Project: *Łukasiewicz logic, MV-algebras and duality*.
- Supervisor: Luca Spada.

Education

PhD · Mathematics *2017.09.25 – 2021.04.09*

University of Milan (Italy), Department of Mathematics.

- Thesis: *On the axiomatisability of the dual of compact ordered spaces*.
- Supervisor: Vincenzo Marra.
- Grade: Excellent with honors.
- Abstract published in the Bulletin of Symbolic Logic. DOI: 10.1017/bsl.2021.54.

MSc · Mathematics *2015.11.01 – 2017.07.19*

University of Milan (Italy), Department of Mathematics.

BSc · Mathematics *2012.07.16 – 2015.10.21*

University of Milan (Italy), Department of Mathematics.

Publications

Publications in international peer-reviewed journals

1. M. Abbadini, F. Guffanti.
   Freely adding one layer of quantifiers to a Boolean doctrine.
   Annals of Pure and Applied Logic. (2027) 178(3):103841. DOI: 10.1016/j.apal.2026.103841.

2. M. Abbadini, L. Spada.
   The unification type of Łukasiewicz logic with a bounded number of variables.
   The Journal of Symbolic Logic. First View, 1–18, 2025. DOI: 10.1017/jsl.2025.10149.

3. M. Abbadini, G. Bezhanishvili, L. Carai.
   On the lack of colimits in various categories arising in pointfree topology and algebraic logic.
   Theory and Applications of Categories. (2026) 45(20):759–778. URL: `http://www.tac.mta.ca/tac/volumes/45/20/45-20abs.html`.

4. M. Abbadini, V. Marra, L. Spada.
   Stone–Gelfand duality for metrically complete lattice-ordered groups.
   Advances in Mathematics. (2025) 461:110067. DOI: 10.1016/j.aim.2024.110067.

5. M. Abbadini, I. Di Liberti.
   Duality for coalgebras for Vietoris and monadicity.
   The Journal of Symbolic Logic. (2025) 90(3):1022–1055. DOI: 10.1017/jsl.2024.14.

6. M. Abbadini, P. Aglianò, S. Fioravanti.
   Varieties of MV-monoids and positive MV-algebras.
   Journal of Algebra. (2025) 677:690–744. DOI: 10.1016/j.jalgebra.2025.04.027.

7. M. Abbadini, F. Guffanti.
   Quantifier-free formulas and quantifier alternation depth in doctrines.
   Journal of Pure and Applied Algebra. (2025) 229(8):108004. DOI: 10.1016/j.jpaa.2025.108004.

8. M. Abbadini, D. Hofmann.
   Barr-coexactness for metric compact Hausdorff spaces.
   Theory and Applications of Categories. (2025) 44(6):196–226. URL: `http://www.tac.mta.ca/tac/volumes/44/6/44-06abs.html`.

9. M. Abbadini, G. Bezhanishvili, L. Carai.
   Vietoris endofunctor for closed relations and its de Vries dual.
   Topology Proceedings. (2024) 64:213–250. URL: `http://topology.nipissingu.ca/tp/reprints/v64`.

10. M. Abbadini, G. Bezhanishvili, L. Carai.
    MacNeille completions of subordination algebras.
    Cahiers de Topologie et Géométrie Différentielle Catégoriques. (2024) 65(2):151–199. URL: https://cahierstgdc.com/wp-content/uploads/2024/04/ABBADINI-Al-LXV-2.pdf.

11. M. Abbadini, G. Bezhanishvili, L. Carai.
    A generalization of de Vries duality to closed relations between compact Hausdorff spaces.
    Topology and its Applications. (2023) 337:108641. DOI: 10.1016/j.topol.2023.108641.

12. M. Abbadini, L. Reggio.
    Barr-exact categories and soft sheaf representations.
    Journal of Pure and Applied Algebra. (2023) 227(12):107413. DOI: 10.1016/j.jpaa.2023.107413.

13. M. Abbadini, P. Jipsen, T. Kroupa, S. Vannucci.
    A finite axiomatization of positive MV-algebras.
    Algebra Universalis. (2022) 83:28. DOI: 10.1007/s00012-022-00776-3.

14. M. Abbadini, L. Spada.
    Are locally finite MV-algebras a variety?
    Journal of Pure and Applied Algebra. (2022) 226(4):106858. DOI: 10.1016/j.jpaa.2021.106858.

15. M. Abbadini.
    Equivalence à la Mundici for commutative lattice-ordered monoids.
    Algebra Universalis. (2021) 82:45. DOI: 10.1007/s00012-021-00736-3.

16. M. Abbadini, L. Reggio.
    On the axiomatisability of the dual of compact ordered spaces.
    Applied Categorical Structures. (2020) 28(6):921–934. DOI: 10.1007/s10485-020-09604-y.

17. M. Abbadini.
    Operations that preserve integrability, and truncated Riesz spaces.
    Forum Mathematicum. (2020) 32(6):1487–1513. DOI: 10.1515/forum-2018-0244.

18. M. Abbadini.
    Dedekind σ-complete ℓ-groups and Riesz spaces as varieties.
    Positivity. (2020) 24(4):1081–1100. DOI: 10.1007/s11117-019-00718-9.

19. M. Abbadini.
    The dual of compact ordered spaces is a variety.
    Theory and Applications of Categories. (2019) 34:1401–1439. URL: http://www.tac.mta.ca/tac/volumes/34/44/34-44abs.html.

Peer-reviewed book chapters

20. M. Abbadini, A. Jung.
    On the symmetry behind duality.
    To appear in *Mai Gehrke on Logic, Algebra, and Duality*, Springer. Preprint at arXiv:2507.18245.

21. M. Abbadini, D. Mundici.
    Unital Specker ℓ-groups and boolean multispaces.

To appear in a volume dedicated to Hilary Priestley, Outstanding Contributions to Logic, Springer. Edited by B. Davey, L. M. Cabrer, A. Palmigiano. Preprint at arXiv:2508.21500.

Proceedings in international peer-reviewed conferences

22. M. Abbadini, F. Di Stefano, L. Spada.
    Unification in Łukasiewicz logic with a finite number of variables.
    In ***Information Processing and Management of Uncertainty in Knowledge-Based Systems***. (2020) 1239:622–633. Springer International Publishing. DOI: 10.1007/978-3-030-50153-2_46.

Preprints

23. M. Abbadini.
    Formal squares over the unit square separate FS-domains from RB-domains.
    Preprint at arXiv:2608.08681.

24. M. Abbadini.
    A topological proof that compact Hausdorff spaces are not finitely co-concrete.
    Preprint at arXiv:2607.29103.

25. M. Abbadini, F. Guffanti.
    The doctrinal Gödel's completeness theorem and the type space functor.
    Preprint at arXiv:2607.21115.

26. M. Abbadini, F. Guffanti.
    On the Beck–Chevalley condition.
    Preprint at arXiv:2607.06386.

27. M. Abbadini, R. N. Almeida, I. Arrieta.
    A topos for étale-finite Heyting algebras.
    Preprint at arXiv:2606.03861.

28. M. Abbadini, A. Přenosil.
    Duality for finitely valued algebras.
    Preprint at arXiv:2505.11490.

Grants and awards

Grants and fellowships

2025–2028 **FSR Incoming Postdoctoral Fellowship.** Awarded by Université catholique de Louvain (Belgium). 3 years.

2023–2025 **Marie Skłodowska–Curie Postdoctoral Fellowship (MSCA PF).** (220,908.48€; 187,096.00£.) Awarded after EU MSCA evaluation; funded by UK Research and Innovation under the Horizon Europe funding guarantee. Host institution: University of Birmingham (UK), School of Computer Science. Scientific Area: Mathematics. Grant EP/Y015029/1. Project "DCPOS".

National qualifications

2024 **Italian national qualification** (Abilitazione Scientifica Nazionale, Seconda Fascia, Competition Sector 01/A1 – Mathematical Logic and Complementary Mathematics).

2023 **French national qualification** as maître de conférences (associate professor) in Mathematics.

2023 **French national qualification** as maître de conférences (associate professor) in Computer Science.

Awards and scholarships

2026 **Winner of one of the two "Paolo Gentilini Prizes".** Awarded by the Italian Association for Logic and its Applications (AILA) to a young researcher in recognition of distinguished contributions to mathematical logic.

2025 **Winner of the "Ada Lettieri Prize".** Awarded by the Italian Association for Logic and its Applications (AILA) for the paper "Stone-Gelfand duality for metrically complete lattice-ordered groups".

2023 **Special mention for the "AILA–UMI Franco Montagna Prize" for PhD theses in mathematical logic and its applications.** Awarded by the Italian Association for Logic and its Applications (AILA) and the Italian Mathematical Union (UMI) for the PhD thesis "On the axiomatisability of the dual of compact ordered spaces".

2023 **Scholarship to conduct one month of research abroad.** Awarded by the Italian National Institute of High Mathematics 'Francesco Severi' (INDAM). Declined, as a Marie Skłodowska–Curie Postdoctoral Fellowship was awarded for the same period.

2017–2020 **PhD scholarship.** Three years, fully funded.

2016/17, Fall **Erasmus scholarship.**

2015/16, Spr. **Erasmus scholarship.**

2012.09.11 **INDAM merit award for outstanding first-year mathematics students.** Awarded by the Italian National Institute of High Mathematics 'Francesco Severi' (INDAM).

Invited talks

1. Workshop on definability, algebra and duality in non-classical logic. University of Amsterdam, Amsterdam (Netherlands). (2026.10.29)
   A topos for étale-finite Heyting algebras.

2. Meeting of the Italian Association of Logic and its Applications. Rome (Italy). Invited as recipient of the "Ada Lettieri 2025" prize and of one of the two "Paolo Gentilini 2026" prizes. (2026.09.15 – 2026.09.18)
   A categorical duality for metrically complete lattice-ordered Abelian groups with unit.

3. 11th International Symposium on Domain Theory and Its Applications. Sichuan University, Chengdu (China). Keynote speaker. (2026.08.06)
   On the symmetry between openness and compactness.

4. Topology, Algebra, and Categories in Logic. Kraków (Poland). Plenary speaker. (2026.07.27)
   On the symmetry behind duality.

5. Conference on Theoretical and Computational Algebra 2026. University of Minho, Guimarães (Portugal). Invited to the Thematic Session on Duality theory. (2026.06.29)
   Adding quantifiers via Stone duality.

6. Workshop on Categorical-Algebraic Structures. Stellenbosch University (South Africa). (2026.04.30)
   Ideals and slices over the initial algebra.

7. International Workshop on Topological Methods in Logic. Ivane Javakhishvili State University (Georgia). (2025.06.25)
   The unification type of Łukasiewicz logic with a bounded number of variables.

8. Association for Symbolic Logic North American Annual Meeting. Iowa State University (USA). Invited to the Special Session on Algebraic Logic. (2024.05.15)
   A duality for metrically complete lattice-ordered groups.

9. Southern and Midlands Logic Seminar. University of Birmingham (UK). (2023.12.13)
   Natural dualities.

10. Third Algebra Week (2023). University of Siena (Italy). (2023.07.05)
    Positive subreducts of MV-algebras.

11. Boolean Algebras, Lattices, Universal Algebra, Set Theory, Topology. University of North Carolina at Charlotte (USA). Plenary speaker. (2023.05.17)
    Positive MV-algebras.

12. Duality, Order, (Co)algebras, Topology and Related topics. Online. (2021.07.07)
    Unital commutative distributive ℓ-monoids and their unit intervals.

13. Third Algebra Week (2020). University of Siena (Italy). (2020.05) (Canceled due to COVID-19 pandemic.)
    Equivalence à la Mundici for lattice-ordered monoids.

Supervision and teaching

Supervision and thesis committees

forthcoming **Incoming member of PhD support committee (*comité d'accompagnement*).** *Université catholique de Louvain (Belgium), Research Institute in Mathematics and Physics.*

- Role: advise and support the PhD student and take part in the confirmation review in the second year of the PhD program.
- PhD Student: Aditya Prakash.
- Thesis project: *Axiomatic Mal'cev Conditions*.

2026.06.22 **Master's thesis reader (*lecteur de mémoire*).** *Université catholique de Louvain (Belgium), School of Mathematics.*

- Role: member of the thesis reading committee, assessing the written mémoire and participating in the thesis evaluation.
- Student: Piero Possidente (University of Turin; Erasmus thesis at Université catholique de Louvain).

2025.09.01 – present **Member of PhD support committee (*comité d'accompagnement*).** *Université catholique de Louvain (Belgium), Research Institute in Mathematics and Physics.*

- Role: advise and support the PhD student and take part in the confirmation review in the second year of the PhD program.
- PhD Student: François Cayphas.

2022.10.24 **Co-supervision of Bachelor's thesis.** *University of Salerno (Italy), Department of Mathematics.*

- Thesis: *Universal constructions in Algebra and in Category Theory*.
- Student: Chiara Carratù.

2021.09.28 **Co-supervision of Bachelor's thesis.** *University of Salerno (Italy), Department of Mathematics.*

- Thesis: *Lawvere's theories: an algebraic approach to Universal Algebra*.
- Student: Maria Bevilacqua.

Teaching

2025/26, Spr. **Course instructor.** *Université catholique de Louvain (Belgium), School of Mathematics.*

- Course: *Categorical Dualities in Logic*.
- Level: MSc in Mathematics.
- Tasks: lectures, office hours, preparation and assessment of oral examinations.
- 22.5 h.

2022/23, Fall **Teaching assistant.** *University of Salerno (Italy), Department of Chemistry and Biology.*

- Course: *Mathematical Methods for Environmental Sciences*.
- Level: BSc in Environmental Sciences.
- Tasks: exercise classes.
- 20 h.

2022/23, Fall **Teaching assistant.** *University of Salerno (Italy), Department of Mathematics.*

- Course: *Mathematical Logic*.
- Level: BSc in Mathematics.
- Tasks: exercise classes.
- 20 h.

2021/22, Fall **Adjunct professor.** *University of Salerno (Italy), Department of Chemistry and Biology.*

- Course: *Mathematics 1*.
- Level: BSc in Environmental Sciences.
- Co-taught with Serafina Lapenta.
- Tasks: classes, office hours, preparation and evaluation of written exams, conducting oral examinations.
- 16 h.

2021/22, Fall **Teaching assistant.** *University of Salerno (Italy), Department of Mathematics.*

- Course: *Mathematical Logic*.
- Level: BSc in Mathematics.
- Tasks: exercise classes.
- 20 h.

2019/20, Fall **Teaching assistant.** *University of Milan (Italy), Department of Mathematics.*

- Course: *Algebra 2*.
- Level: BSc in Mathematics.
- Tasks: exercise classes, office hours, exam assistant.
- 35 h.

Additionally, I completed the Italian **teacher-training program "24 CFU"**: 24 ECTS (university credits), covering didactic methods and technologies, educational psychology and pedagogy (University of Milan, 2019/20).

Student-oriented seminars and mini-courses

Seminar for PhD students and postdoctoral researchers in Computer Science.
University of Birmingham (UK), School of Computer Science. (2025.02.25)
The algebras of classical first-order logic.

Seminar for PhD students and postdoctoral researchers in Computer Science.
University of Birmingham (UK), School of Computer Science. (2025.02.11)
The sequent calculus whose semantics admits the empty structure.

Mini-course (3h total) for PhD students and postdoctoral researchers in Computer Science.
University of Birmingham (UK), School of Computer Science. (2024.10.17 and 2024.10.24)
Stone duality.

Seminar for graduate students in Mathematics.
New Mexico State University (USA), online. (2022.03.24)
Lawvere's algebraic theories.

Seminar for Master students in Mathematics.
University of Salerno (Italy). (2021.05.28)
The algebraic approach to Constraint Satisfaction Problems (CSP).

Seminar for Bachelor students in Mathematics.
University of Milan (Italy). (2019.12.05)
The Rubik's Cube. Ideas to create a solution strategy.

Participation in research projects

DCPOS: Dualities via ComPactness, Openness and their Symmetry *2023.09 – 2025.08*

University of Birmingham (UK)

Role: ***Experienced Researcher***.

Marie Skłodowska–Curie Postdoctoral Fellowship (MSCA PF), funded by UK Research and Innovation under the Horizon Europe funding guarantee. Grant EP/Y015029/1.

MOSAIC: Modalities in Substructural Logics: Theory, Methods and Applications *2021.12 – 2023.09*

University of Salerno (Italy)

Role: ***Team member***.

Rise-MSCA-2020, funded by the European Union. Project 101007627.

Theory and applications of resource sensitive logics *2019.12 – 2022.12*

University of Salerno (Italy)

Role: ***Team member***.

PRIN, funded by the Italian Ministry of University and Research. Project 20173WKCM5.

Organization of conferences and seminars

International conference "Peripatetic Seminar on Sheaves and Logic 113" *2026.09.26 – 2026.09.27*

Université catholique de Louvain (Belgium)

Role: ***member of the organizing committee***.

International conference.

International workshop "6th Workshop on Categorical Algebra: recent developments and future perspectives" *2026.07.01 – 2026.07.04*

Mountain University of the University of Milan (UniMont)

Role: ***member of the organizing committee***.

International workshop.

Achim Jung's mini-course on *denotational semantics* *Autumn 2024*

University of Birmingham (UK), School of Computer Science

Role: ***initiator and organizer***.

Coordinated the schedule with the lecturer and the research group, arranged rooms and live streaming, and publicised the sessions within the department.

Series of seminars in logic *2023.04.17 – 2023.06.17*

University of Salerno (Italy), Department of Mathematics

Role: ***organizer***.

6 seminars + 1 final workshop.

International conference "LATD 2022 and MOSAIC 2022 Kick Off Meeting" *2022.09.05 – 2022.09.10*

Paestum (Italy)

Role: ***member of the organizing committee***.

International conference (84 participants).

Workshop "PhD^{2}ays 2020" *2020.12.09 – 2020.12.11*

University of Milan (Italy), Online

Role: ***member of the organizing committee***.

Local workshop, in which PhD students present their research to postgraduate students (∼ 35 participants).

Visits

Long secondments (≥ 1 month)

2026 **University of Cape Town (South Africa).** Visiting G. Janelidze. 1 month.

2023 **University of the Witwatersrand (South Africa).** Visiting C. van Alten. 1 month.

2022 **Chapman University (USA).** Visiting M. A. Moshier. 2 months.

Short research visits (< 1 month)

2025 **Czech Academy of Sciences (Czech Republic).** Visiting A. Přenosil and W. Poiger. 5 days.

2025 **Université catholique de Louvain (Belgium).** Visiting M. Gran. 4 days.

2024 **University of Aveiro (Portugal).** Visiting D. Hofmann. 5 days.

2024 **University of Luxembourg (Luxembourg).** Visiting F. Guffanti. 3 days.

2024 **Université catholique de Louvain (Belgium).** Visiting M. Gran. 2 days.

2023 **University of Bern (Switzerland).** Visiting G. Metcalfe. 3 days.

2023 **University of Aveiro (Portugal).** Visiting D. Hofmann. 8 days.

2022 **Université Paris Cité (France).** Visiting S. van Gool. 3 days.

2022 **Oxford University (UK).** Visiting L. Reggio. 5 days.

2020 **University of Salerno (Italy).** Visiting L. Spada. 4 days.

2019 **University of Nice (France).** Visiting W. Fussner. 3 days.

2019 **Czech Academy of Sciences (Czech Republic).** Visiting L. Reggio. 4 days.

2018 **University of Salerno (Italy).** Visiting L. Spada. 6 days.

Visits hosted

2026 **L. Spada.** At Université catholique de Louvain (Belgium). 2 weeks.

2025 **L. Spada.** At Université catholique de Louvain (Belgium). 4 days.

2025 **R. N. Almeida.** At the University of Birmingham (UK). 2 weeks.

2024 **S. van Gool.** At the University of Birmingham (UK). 5 days.

2023 **A. Přenosil.** At the University of Salerno (Italy). 5 days.

2023 **F. Guffanti.** At the University of Salerno (Italy). 2 days.

Outreach

Interview published by *AILA x MaddMaths!*. *Dare forma alla logica: formule, algebre e spazi – Intervista a Marco Abbadini.* Italian original. English version. (2026.07.22)

Outreach blog post on *Maddmaths!* (Italian outreach platform in mathematics) about Stone duality and its recent developments. M. Abbadini, *Attraverso lo specchio.* Italian original. English version. (2025.05.06)

Organizer and volunteer in educational events for the general public, illustrating through games the mathematical mechanisms of gambling and the systematic disadvantage for the player. *Slotmob* (2015.06.05) and *Apericena d'azzardo* (2014.04.11).

Additional talks

Contributed talks

14. Workshop on General Algebra. University of Ljubljana (Slovenia). (2026.06.05)
    Ideals and slices over the initial algebra.

15. Peripatetic Seminar on Sheaves and Logic. University of Nottingham (UK). (2026.03.29)
    The doctrinal Herbrand's theorem and its Stone dual.

16. ItaCa Workshop. University of Milan (Italy). (2025.12.22)
    The doctrinal Herbrand's theorem and its Stone dual.

17. Australasian Association for Logic Conference. University of Queensland (Australia, online). (2025.11.03)
    Natural duality for finitely valued algebras.

18. Category Theory Octoberfest. Online. (2025.10.25)
    On the category of metric compact Hausdorff spaces.

19. Logic, Algebra and Truth Degrees. University of Siena (Italy). (2025.07.25)
    The algebraic structure of spaces of integrable functions.

20. Logic Colloquium. TU Wien (Austria). (2025.07.07)
    An algebraic version of Herbrand's theorem.

21. Workshop on General Algebra. Bern (Switzerland). (2025.06.21)
    The algebraic structure of spaces of integrable functions.

22. Peripatetic Seminar on Sheaves and Logic. Tallinn University of Technology (Estonia). (2025.05.03)
    Barr-coexactness of metric compact Hausdorff spaces.

23. Meeting of the Italian Association of Logic and its Applications. University of Udine (Italy). (2024.09.04)
    Vietoris endofunctor for closed relations and its de Vries dual.

24. Summer Conference on Topology and its Applications. University of Coimbra (Portugal). (2024.07.09)
    Vietoris endofunctor for closed relations and its de Vries dual.

25. Topology, Algebra, and Categories in Logic. University of Barcelona (Spain). (2024.07.04)
    Vietoris endofunctor for closed relations and its de Vries dual.

26. Workshop on General Algebra. South-West University "Neofit Rilsk" (Bulgaria). (2024.02.10)
    Natural dualities with an infinite dualizing object.

27. Modalities in Substructural Logics: Theory, Methods and Applications. Vienna (Austria). (2023.09.29)
    Stone duality for finitely valued algebras with a near-unanimity term.

28. Peripatetic Seminar on Sheaves and Logic. University of Palermo (Italy). (2023.09.17)
    Soft sheaf representations in Barr-exact categories.

29. Logic, Algebra and Truth Degrees. Ivane Javakhishvili State University (Georgia). (2023.09.11)
    Free extensions.

30. Logic Colloquium. University of Milan (Italy). (2023.06.08)
    Jónsson-Tarski duality beyond dimension 0.

31. ItaCa Workshop. University of Pisa (Italy). (2022.12.21)
    Comonadicity over Set *of coalgebras of Vietoris functors*.

32. Meeting of the Italian Association of Logic and its Applications. Vanvitelli University (Italy). (2022.09.13)
    Duality for metrically complete Abelian ℓ-groups.

33. Topology, Algebra and Categories in Logic. University of Coimbra (Portugal). (2022.06.24)
    An approach à la de Vries for compact Hausdorff spaces and closed relations.

34. Category Theory. University of Genoa (Italy). (2021.09.01)
    The opposite of the category of compact ordered spaces is monadic over the category of sets.

35. Boolean Algebras, Lattices, Universal Algebra, Set Theory, Topology. New Mexico State University (USA, online). (2021.06.13)
    The opposite of the category of compact ordered spaces as an infinitary variety.

36. Workshop on General Algebra. University of Siena (Italy). (2020.02.21)
    Priestley duality above dimension zero: algebraic axiomatisability of the dual of compact ordered spaces.

37. Topology, Algebra, and Categories in Logic. University of Nice (France). (2019.06.20)
    Norm-complete Abelian ℓ-groups: equational axiomatization.

38. PhDs in Logic. University of Bern (Switzerland). (2019.04.25)
    Stone-Gelfand duality for groups.

39. Boolean Algebras, Lattices, Universal Algebra, Set Theory, Topology. University of Denver (USA). (2018.08.10)
    Operations that preserve integrability, and truncated Riesz spaces.

Talks at local seminars

40. University of KwaZulu-Natal (Durban, South Africa). (2026.05.21)
    On the symmetry between openness and compactness.

41. Université catholique de Louvain (Belgium). (2025.10.29)
    Bridging Ordered Groups and Topology: Representation Theorems and Categorical Dualities.

42. University of Birmingham (UK). (2025.06.05)
    On de Groot and Lawson self-dualities.

43. University of Birmingham (UK). (2025.05.08)
    Freely adding a layer of Heyting implication to a distributive lattice.

44. University of Salerno (Italy). (2025.04.24)
    An algebraic version of Herbrand's theorem.

45. Université catholique de Louvain (Belgium). (2025.01.07)
    Coalgebraic flavour of metric compact Hausdorff spaces.

46. University of Milan (Italy). (2024.10.31)
    Quantifier-free fragments and quantifier alternation depth in doctrines.

47. University of Manchester (UK). (2024.10.09)
    Quantifier-free fragments and quantifier alternation depth in doctrines.

48. University of Padua (Italy). (2024.05.02)
    Quantifier alternation depth in universal Boolean doctrines.

49. University of Birmingham (UK). (2023.11.23)
    An abstraction of the unit interval with Euclidean topology and denominators.

50. Université catholique de Louvain (Belgium). (2023.11.13)
    Soft sheaf representations in Barr-exact categories.

51. University of Bern (Switzerland). (2023.04.20)
    Positive MV-algebras.

52. University of the Witwatersrand (South Africa). (2023.03.30)
    Unique embeddability property.

53. Paris-Saclay University (France, online). (2023.01.04)
    An abstraction of the unit interval with denominators.

54. University of Padua (Italy, online). (2022.12.05)
    Dualities for compact Hausdorff spaces.

55. Université Paris Cité (France). (2022.11.25)
    A generalization of de Vries duality to closed relations between compact Hausdorff spaces.

56. University of Amsterdam (Netherlands). (2022.05.11)
    A generalization of de Vries duality to closed relations between compact Hausdorff spaces.

57. New Mexico State University (USA, online). (2022.03.14)
    Free extension for universal algebras.

58. University of Caserta (Italy). (2022.02.14)
    Positive MV-algebras.

59. University of Denver (USA, online). (2021.05.05)
    Is the category of locally finite MV-algebras equivalent to an equational class?

60. University of Denver (USA, online). (2020.05.08)
    Equivalence à la Mundici for lattice-ordered monoids.

61. University of Salerno (Italy). (2020.01.23)
    Equivalence à la Mundici for lattice-ordered monoids.

62. University of Nice (France). (2019.10.03)
    On concrete dual adjunctions.

63. Czech Academy of Sciences (Czech Republic). (2019.02.06)
    The dual of compact ordered spaces is a variety.

64. University of Salerno (Italy). (2018.12.13)
    Archimedean Cauchy-complete MV-algebras form a variety.

65. University of Milan (Italy). (2018.07.06)
    Operations that preserve integrability: characterization and related algebraic structures.

Other talks and posters

66. Autumn school “Proof and Computation”, Fischbachau (Germany). 5-minute lightning talk. (2024.09.19)
    Algebras of logic, step-by-step.

67. Congresso dell’Unione Matematica Italiana, University of Pavia (Italy). Poster. (2019.09.02)
    Dualità di Stone-Gelfand per i gruppi.

Professional memberships

Member of the Italian Association for Logic and its Applications (AILA).
Member of the Marie Curie Alumni Association (MCAA).

Languages

Italian	**C2**. (Mother tongue.)
English	**C1**.
French	**B2**. (B2 course completed at UCLouvain, 2025/26; C1 course in progress 2025/26.)
Spanish	**B2**.
German	**A2**. (2016/10/12: B2 certificate issued by University of Stuttgart, Germany.)